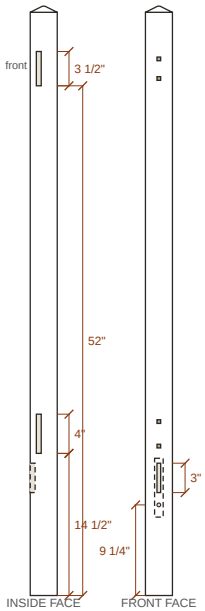
Queen bed frame: joinery

Inches. Heights from the floor; post and leg positions from their front faces (toward the foot). Details drawn to different scales: work from the numbers. Overall plan and cut list: bed-frame-drawing.svg.

A. HEAD POST (2)

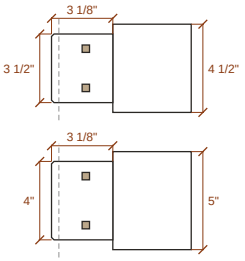
2 3/4" square, 60 1/8" long with its cap.
Left post shown; the right is its mirror.



Through-mortises 1/2" x 3 1/2" and 1/2" x 4", 5/8" from the front face, cut halfway from the inside and outside faces so neither blows out.
Pegs 3/8" square, 1 1/2" deep from the front face, centered across it, at 15 1/4", 17 3/4", 52 3/4", 54 3/4" up.
Stub mortise for the side rail: 3/8" x 3" x 9/16" deep, centered, 10 1/2" up.
Bed bolt: 3/8" hole through, front to back, 9 1/4" up, centered;
1" counterbore 1/2" deep on the back face for the bolt head.
Pyramid cap: 5/8" rise to a 1/4" flat; plane or sand it before glue-up.

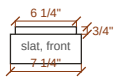
B. HEADBOARD RAILS: THROUGH-TENONS

1/2" thick tenons, centered on the 1 1/4" rail; shoulders 1/2" top and bottom.
The dashed line is the post's outside face: the tenon stands 3/8" proud, its end chamfered 1/8".



C. HEADBOARD SLATS (7) AND THEIR MORTISES

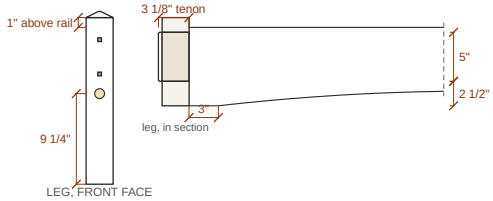
Slats 3/4" x 7 1/4" x 34" with tenons (32 1/2" between the rails).
Stub tenons 3/8" thick x 3/4" long, 1/2" shoulders each side; flush with the rails' front faces.



Mortises in both rails, 3/8" wide x 13/16" deep, 3/16" from the rails' front faces; measured from the rail's shoulder (the post's inside face):
slat 1: 1 1/2" to 7 3/4"
slat 2: 9 3/4" to 15 15/16"
slat 3: 17 15/16" to 24 3/16"
slat 4: 26 3/16" to 32 7/16"
slat 5: 34 7/16" to 40 11/16"
slat 6: 42 11/16" to 48 7/8"
slat 7: 50 7/8" to 57 1/8"

D. FOOT LEGS (2) AND FOOT RAIL

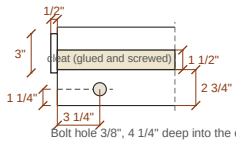
Legs 2 3/4" square x 17 5/8" with the cap; the rail 7/8" thick, 8" tall at its ends, its bottom arched 1 1/2".
Left leg and the rail's left end shown; the right is the mirror.



Rail 58 5/8" long between the legs, 64 7/8" overall with its tenons.
Through-tenons 3/8" thick, centered on the rail; mortises 1 3/16" from the legs' front faces.
The arch: a fair curve from 3" in from each leg, 1 1/2" high at the middle (29 5/16" from the shoulder). Bend a thin batten through the three points and trace it.
Bed bolt from the front face, 9 1/4" up, 1" counterbore 1/2" deep, brass cover;
stub mortise for the side rail on the back face, as on the head posts.
Pegs 3/8" square through the front face, centered, at 11 1/4" and 14 3/4" up.

E. SIDE RAILS (2): ENDS

7/8" x 6", 79 9/16" between the posts and legs, 80 9/16" overall with the stub tenons.
The inside face shown (the end at a post: both ends the same).



Bolt hole 3/8", 4 1/4" deep into the end grain, 1 1/4" up, centered.
Nut hole 1" Forstner from the inside face, 3/4" deep, 3 1/4" from the end:
check it against the bolt before drilling.
Stub tenon 3/8" thick, 3" tall, 1/2" long, centered.
Cleats run the rail's full length, above the nut hole; each slat screwed to them with one #8 x 1 1/4" at each end, and to the center rail with one.

Updated November 10, 2026